

Pritam Kumar Panda

Computational Biophysicist · AI-Driven Drug Discovery · Agentic Scientific Systems
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PROFILE

Computational scientist bridging quantum-scale physics, structural biology, and machine learning to compress drug-discovery timelines. Builds production GPU/HPC platforms and agentic AI tooling that move programs from screen to clinic. H-index 49 · 6,300+ citations · 89 Q1 publications · 1 U.S. patent.

WORK EXPERIENCE

Stanford University

Jan 2025 – Present

Postdoctoral Scholar — Anesthesiology, Perioperative & Pain Medicine

Palo Alto, CA

- **Advanced a DoD/DARPA anesthesia program to Phase II clinical trials** by architecting an AI-driven drug discovery platform that screened 2.5B small molecules and generated 100,000+ AlphaFold-based receptor conformations.
- **Cut experimental candidate testing 60%** by implementing GPU-accelerated atomistic simulation workflows coupled with active-learning docking triage and ADMET/QSAR profiling.

STACK Python; PyTorch/JAX; RDKit; AlphaFold2/3; Schrödinger; GROMACS; OpenMM; AutoDock-GPU; Vina; Glide; CDOCKER; Boltz2; AWS/HPC; CUDA

German Cancer Research Center (DKFZ)

Jul 2023 – Dec 2024

Data Manager / Bioinformatician

Heidelberg, Germany

- **Resolved ~20 client-facing technical issues daily** and improved pipeline reliability by designing reproducible multi-omics workflows across HPC for WES/WGS/RNA-seq/scRNA-seq/ATAC-seq.
- **Standardized pipeline validation and client onboarding across teams** by leading development of the HI-TRON FAIR data portal and authoring SOPs.

STACK Nextflow; Snakemake; Bash; Docker/Singularity; HPC/SLURM; Vue.js; R; Python; GitLab/GitHub; CI/CD

Karolinska Institute

Mar 2021 – Dec 2022

Affiliated Researcher

Stockholm, Sweden

- **Produced 2 peer-reviewed publications and 1 U.S. patent** by developing predictive clinical ML models for patient stratification from genomic and antibody-profiling data.
- **Enabled 3 ongoing clinical trials** by engineering immunoinformatics pipelines processing 2TB+ multimodal clinical datasets with startup initiative Inflonava AB, Sweden.

STACK Python; R; scikit-learn; immunoinformatics pipelines; clinical genomics & antibody-profiling data

Uppsala University

Nov 2018 – Mar 2023

PhD Researcher — Computational Biophysics & Quantum Physics

Uppsala, Sweden

- **Originated GENOME2QUNOME, the first quantum-to-biological modeling framework**, bridging DFT-based 2D-material simulation with atomistic modeling for nanopore sensing and bio-inspired materials.
- **Informed material selection for an industry partner (Hitachi ABB)** by leading ab initio electronic and thermal-transport studies on cellulose-based transformer materials.

STACK VASP; SIESTA/TranSIESTA; DFT; Python; ML-augmented electronic-structure modeling

Freiburg Medical Center

Sep 2017 – Oct 2018

NGS Data Analyst — Pediatric Hematology & Oncology

Freiburg, Germany

- **Cut manual runtime 70% and improved alignment speed 2.5×** by engineering high-throughput WES/WGS variant-calling pipelines processing 200+ clinical samples/month with HPC parallelization.
- **Contributed to a Nature Medicine publication on clonal hematopoiesis trajectories** by co-authoring analysis with the European MDS Working Group.

STACK BWA-MEM; STAR; GATK; ANNOVAR/VEP/CADD; HPC/SLURM; RNA-seq

ADVISORY & LEADERSHIP

Revilico, Inc.

Scientific Advisor

2025 – Present

Remote

- Architected Revilico's core computational drug-discovery platform — GPU-accelerated, production-grade simulation infrastructure uniting docking, FEP, GROMACS, ADMET/QSAR, NVIDIA Parabricks, multi-omics, and quantum-chemistry workflows into a single operating system for drug discovery, built from the ground up with Claude-based agentic tooling.

Colgate-Palmolive

2021 – 2023

Bioinformatics Consultant

Remote

- Co-developed a CADD synergy-based drug-design pipeline for photoaging and hyperpigmentation, yielding a peer-reviewed publication; authored SOPs for incoming scientists.

Nerdalytics

2021 – 2023

Founder & CEO

Uppsala, Sweden

- Founded and led a bioinformatics consultancy for the pharmaceutical industry, owning strategy, product, and client delivery. Organized two international longevity hackathons, mentoring 160 researchers across 10 countries with LongHack/VitaDAO.

OPEN - SOURCE PROJECTS

- **BioClaw** (2026) Claude-powered ReAct agent exposing 17 bioinformatics tools via CLI, REST API, and Next.js UI.
- **Virtual-Cell IC50 Model** (2026) Trained on 8.8M protein–ligand complexes (SAIR) to predict IC50 across CCLE and Human Protein Atlas lines.
- **Automated FEP Pipeline** (2026) One-command FEP + MD + metadynamics workflow for protein–ligand binding free-energy prediction.
- **nf-core/moleculardocking & /moleculardynamics** (2026) Production-grade Nextflow pipelines for virtual screening and MD, adopted for reproducible HPC execution.
- **PandaDock** (2025) SE(3)-equivariant graph neural network docking platform; R > 0.8 vs. experimental binding affinities.
- **PandaMap** (2025) Interactive protein–ligand visualization integrating 2D ligand representations with 3D docking outputs.

EDUCATION

Uppsala University · PhD, Physics (Computational Biophysics, Quantum Physics)

Uppsala, Sweden · 2023

D.Y. Patil University · M.Tech, Bioinformatics

Mumbai, India · 2016

Utkal University · M.Sc, Bioinformatics

Bhubaneswar, India · 2014

SKILLS & TECHNOLOGIES

Skills Agentic systems; LLM tool-calling; evals & failure-case curation; dataset curation; genomics; variant calling; structural biology; protein design; molecular docking; molecular dynamics; free-energy perturbation (FEP); workflow orchestration; database & API design; multi-GPU computing; performance profiling.

Technologies Claude Code; ReAct agents; PyTorch; JAX; Hugging Face; scikit-learn; WES/WGS; RNA-seq; scRNA-seq; Nextflow; Snakemake; AWS (S3/EC2/HealthOmics); HPC/SLURM; Docker; Singularity; AlphaFold2/3; RFdiffusion; Boltz2; AutoDock-GPU; Vina; Glide; GROMACS; OpenMM; RDKit; ChimeraX; REST; Next.js; CUDA.

Certifications NVIDIA Foundation Models in Digital Biology; Machine Learning Specialization; Google MLOps for Generative AI (Vertex AI); AWS Cloud Practitioner; Google Business Intelligence.

RECOGNITION & PUBLICATIONS

Publications H-index 49; 6,300+ citations; 89 Q1 publications including Science Advances, Nature Medicine, and Leukemia.

Recognition Elected Member, Sigma Xi; Nextflow Ambassador; Stanford Best Poster Prize (2026); 100+ peer reviews across 30+ journals; 18,000+ LinkedIn followers.

Speaking Invited speaker and presenter at 5+ international summits (Helmholtz, HI-TRON, ASH, EWOG-MDS, Curious Future Insight).

Interests Calisthenics; meditation; gardening; teaching; scientific content creation.